# SECTIONS OF THE HOM COMPLEX COMPUTE DERIVED HOMS INTO K-INJECTIVES

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

*Allowed standard prerequisites (editorial).*

- Restriction of K-injective complexes. Cohomology of Sheaves, Lemma cohomology-lemma-restrict-K-injective-to-open. I^ • |_U is K-injective.

*Source reference.* Original equation 0A8L: sources/stacks-chow/tags.txt, source lines 12919–12919. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 08BS: sources/stacks-chow/tags.txt, source lines 10577–10577. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $L$ and $M$ be objects of $D(\mathcal{O}_X)$. Let $\mathcal{I}^\bullet$ be a K-injective complex of $\mathcal{O}_X$-modules representing $M$. Let $\mathcal{L}^\bullet$ be a complex of $\mathcal{O}_X$-modules representing $L$. Then*

$$H^0(\Gamma(U, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet))) = \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U)$$

*for all $U \subset X$ open.*

*Proof.* We have

$$\begin{aligned} H^0(\Gamma(U, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet))) &= \mathrm{Hom}_{K(\mathcal{O}_U)}(\mathcal{L}^\bullet|_U, \mathcal{I}^\bullet|_U) \\ &= \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U) \end{aligned}$$

The first equality is (0A8L). The second equality is true because $\mathcal{I}^\bullet|_U$ is K-injective by Lemma 08BS. □

*Date*: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.